Technology Stack

Filled project document for GitHub and SkillWallet submission

Date	28 June 2026	Team ID	To be updated in SkillWallet
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction	Team Members	Devireddy Prasanna, Chokka Abitha, Navya Borugadda, Bodi Varshitha, Mulakala Harika Venkata Sridevi

Components and Technologies

Component	Description	Technology
User Interface	Weather input form, dashboard, prediction result page	HTML, CSS, JavaScript
Application Layer	Routes, validation, model loading, prediction response	Python Flask, joblib
Machine Learning	Classification models and evaluation	scikit-learn, XGBoost
Data Processing	Feature cleaning, scaling, splitting, plotting	NumPy, pandas, Matplotlib
Model Artifacts	Saved best model and preprocessing object	model.pkl, scaler.pkl
Development	Notebook and local testing	Jupyter Notebook / Anaconda
Version Control	Project source and deliverables	Git, GitHub
Cloud	Potential public deployment	IBM Cloud

Application Characteristics

Characteristic	Implementation
Open-source frameworks	Flask, scikit-learn, XGBoost, pandas, NumPy, Matplotlib.
Security	Input validation, controlled file loading, no user access to server internals.
Scalability	Separated presentation, application, ML, and data layers for cloud deployment.
Availability	Can run locally for demo and on IBM Cloud for public/mentor access.
Performance	Saved model inference avoids retraining during each prediction.